AS · Unit 1 · WBI11 Molecules, Diet, Transport and Health

Mammalian transport

Mammals are large and very active, so diffusion alone cannot supply their cells. This page explains why a heart and double circulation are needed, how arteries, veins and capillaries are built, how the heart works through the cardiac cycle, and how haemoglobin carries oxygen and carbon dioxide.

SPEC 1.6 Why many animals need a heart and circulation

WHAT THIS SPEC POINT ASKS

Understand why many animals have a heart and circulation: mass transport overcomes the limitations of diffusion in meeting an organism's needs.

The problem: diffusion is only fast over short distances

All cells need a steady supply of O_2 and glucose, and must get rid of CO_2 and urea. A small organism can rely on diffusion. A large one can't, for two reasons:

1. **Distance**: many cell layers mean long diffusion distances, and the time diffusion takes rises steeply with distance.
2. **Surface area : volume ratio falls as size increases**, so the surface is too small to supply the whole volume. Large, active animals also have a **high metabolic rate** (high demand).

WORKED EXAMPLE

Worked example: SA : V

A 1 cm cube: SA = 6 cm², V = 1 cm³ → **6 : 1**.

A 3 cm cube: SA = 6 × 9 = 54 cm², V = 27 cm³ → **2 : 1**.

Tripling the length cut the SA:V ratio to a third, so there is far less surface per unit of volume to supply.

Other shapes: **cylinder** (a vessel, a worm) V = $\pi r^2 l$, curved SA = $2\pi rl$; **sphere** V = $\frac{4}{3}\pi r^3$, SA = $4\pi r^2$.

Use the **radius** (half the diameter): using the diameter is the commonest error. Give the ratio as **x : 1**.

The solution: mass transport

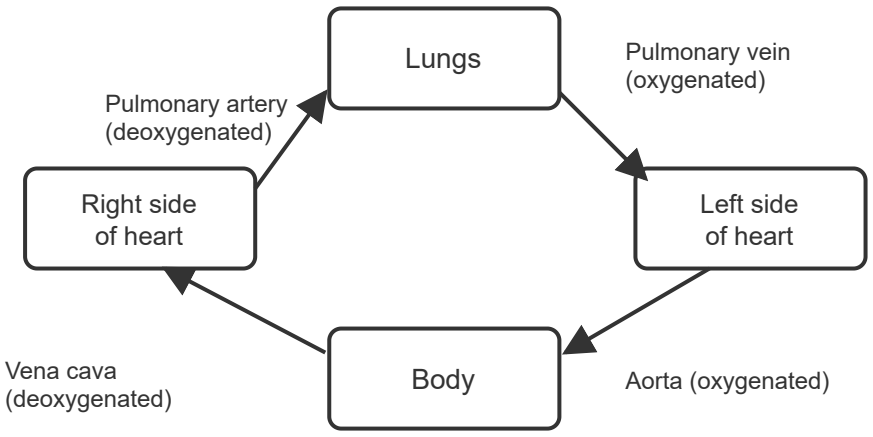
- **Mass transport** (mass flow) is the **bulk** movement of a fluid in one direction, driven by a **pressure difference** made by a pump (the **heart**), inside **vessels**.

- It carries substances quickly between exchange surfaces (lungs, gut) and every tissue, so the final step into or out of each cell is only a short diffusion distance.
- It also **maintains concentration gradients** at exchange surfaces and in tissues, by constantly bringing new blood and carrying away what has diffused in.

Single vs double circulation

	Single (e.g. fish)	**Double** (mammals)
Route	heart → gills → body → heart	heart → lungs → heart → body → heart
Passes through heart per circuit	once	**twice**
Pressure to the body	low (pressure is lost in the gill capillaries)	**high**: blood is re-pressurised after the lungs

Advantages of a double circulation: **low pressure to the lungs** (doesn't damage the delicate capillaries; slower flow allows time for gas exchange) and **high pressure to the body** (fast delivery for an active, endothermic animal). The **septum** keeps oxygenated and deoxygenated blood apart.



⚠ **Beyond the spec · not part of the Edexcel IAL specification**

Examined in past papers even though the spec doesn't list it.

Other animals' circulations. Fish: pressure falls after the gill capillaries, so flow to the body is slow; that is enough because fish aren't endothermic and have a **lower metabolic rate**. Amphibians (WBI11 June 2025): a **three-chambered heart** (two atria, one ventricle, no septum), so some mixing of oxygenated and deoxygenated blood. That is enough for a small animal with a **low O_2 demand**, which also exchanges gases through its moist skin. As in mammals, oxygenated blood from the lungs returns to the **left** atrium. Insects (WBI11 Oct 2021): an **open** circulation, with blood in body cavities rather than vessels; O_2 reaches cells through tracheoles, so oxygenated and deoxygenated blood don't need to be kept apart.

MARK-SCHEME WORDING

- Mass transport: "all the components (of the blood) flow together", "down a pressure gradient", "to overcome the limitations of diffusion".
- Diffusion is "too slow" over large distances. Quote data if given (e.g. 0.5 ms over 1 μm, but 50 ms over 10 μm).
- Organisms with no circulation: every cell is "not far from" the environment, so "diffusion can supply" enough; "low metabolic rate / oxygen demand"; large SA:V.
- A big heart in context (e.g. a giraffe): blood "pumped under high pressure" to reach the head "against gravity".

Often asked as: explain in a named organism (giraffe, butterfly, yak, hydra, frog), define mass transport, calculate SA:V.

EXAM TIP

Name **both** reasons: diffusion distance too long **and** SA:V too small for a high demand. Say what is transported and where, e.g. "O_2 and glucose to respiring cells".

COMMON MISTAKE

Writing about SA:V when the question is about **distance** or **pressure** (e.g. a giraffe's heart). Answer the factor the question points to.

Vague "nutrients and waste": name O_2, glucose and CO_2, and say **where** the concentration gradient is.

In a comparison, describe both animals, not just the mammal.

SPEC 1.7 Blood vessels: structure related to function

WHAT THIS SPEC POINT ASKS

Understand how the structures of capillaries, arteries and veins relate to their functions.

Feature	Artery	Vein	Capillary
Job	carry blood **away** from the heart at **high pressure**	return blood **to** the heart at **low pressure**	**exchange** between blood and tissues

Feature	Artery	Vein	Capillary
Wall	**thick**: lots of **elastic tissue** and **smooth muscle**, collagen outside	**thin**: little elastic tissue or muscle	**one cell thick** (endothelium only)
Lumen	**narrow** relative to wall	**wide**	about the width of one red blood cell
Valves	no (only where arteries leave the heart)	**yes**	no
Pulse	yes	no	no

Memory aid: **A**rteries go **A**way; veins go **in**.

Why each feature matters

- **Elastic tissue** (artery): **stretches** when the ventricle forces blood in, then **recoils** in diastole. This **smooths out pressure surges** and keeps pressure up between beats.
- **Smooth muscle**: **contracts** or relaxes to narrow or widen the lumen, controlling blood flow to different organs.
- **Collagen**: strength, so the wall withstands high pressure and doesn't overstretch or burst.
- **Smooth, folded endothelium**: reduces friction; folds let the artery stretch.
- **Wide lumen in veins**: less resistance at low pressure. **Valves** stop **backflow**. Contracting **skeletal muscles** squeeze veins and push blood towards the heart.
- **Capillary wall one cell thick** → **short diffusion distance**. **Narrow lumen** → red cells pass in single file, pressed against the wall (short distance) and moving **slowly** (more time for exchange). Huge **branched network** → large surface area, and every cell is close to a capillary. **Gaps between cells** let plasma out to form tissue fluid and let white cells squeeze through.

MARK-SCHEME WORDING

- Elastic fibres "stretch" when pressure is high, "widening the lumen", then "elastic recoil" "maintains the high blood pressure" and smooths the flow. In "why doesn't it burst?" questions, recoil is ignored: use stretch + "collagen for (tensile) strength".
- Endothelium: "smooth / flattened", "to reduce friction". Smooth muscle: contracts to control flow (vasoconstriction).
- Capillaries: one cell thick → short diffusion distance for gases; pores let plasma and white cells out; narrow, so red cells are close to the wall.
- Veins: valves "prevent backflow" because pressure is low.
- Artery definition: "carries blood away from the heart". Not "carries oxygenated blood" (the pulmonary artery doesn't).
- Drawing an artery TS: lumen + three layers, labelled tunica intima (endothelium), tunica media (smooth muscle + elastic fibres), tunica externa (collagen). Descriptions instead of names are fine. Label lines must touch the layer, and a wrong label cancels a right one.

- Velocity graphs: flow "decreases through the arterioles", is "low in the capillaries" (huge total cross-sectional area), "increases in the venules". Explain velocity, not pressure.

Often asked as: draw and label a TS of an artery or vein, explain a named feature, read velocity or pressure graphs, calculate a lumen area or mean arterial pressure.

COMMON MISTAKE

"Arteries have thick walls *to pump blood*." Arteries don't pump; the heart does. The wall **withstands** high pressure and the elastic tissue **recoils** to maintain it. Don't swap the verbs: muscle **contracts**, elastic tissue **stretches and recoils**. Capillary walls are made of cells; they don't have "cell walls".

Writing "walls" when you mean "elastic fibres", or mixing up elastic fibres and smooth muscle.

Vein valves drawn the wrong way round: they open in the direction of flow, towards the heart.

"Glucose diffuses out through the pores": examiners only credit diffusion across the wall for gases. Plasma (carrying glucose) is forced out through the gaps.

Forgetting to multiply by the number of vessels in a surface-area calculation.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Tissue fluid: at the arterial end of a capillary, **hydrostatic pressure** is higher than the opposing pull of the blood's water potential, so plasma (minus large proteins and cells) is forced out. At the venous end, hydrostatic pressure has fallen, so water moves back in by **osmosis** (plasma proteins keep the blood's water potential low). Excess fluid drains into **lymph** vessels. High blood pressure → more tissue fluid forms (oedema).

Mean arterial pressure (formula given): MAP = diastolic + ⅓(systolic − diastolic). For 120/80 mmHg: 80 + 40 ÷ 3 = **93 mmHg** (WBI11 Jun 2025).

Lumen area: lumen diameter = outer diameter − 2 × wall thickness; halve it for the radius, then πr^2. Convert units first (e.g. mm → μm) (WBI11 Jun 2025).

Net movement of fluid across a capillary

1. At the arterial end, hydrostatic pressure is greater than the oncotic pressure.

   The blood has just been pumped out of the heart, so it is at high pressure.

2. The net pressure is outwards, so fluid is forced into the tissues.

3. Along the capillary, hydrostatic pressure falls.

   Friction and fluid loss lower the pressure as the blood moves along.

4. At the venous end, oncotic pressure is greater than hydrostatic pressure, so the net pressure is inwards.

5. Fluid returns to the capillary, and the excess drains into lymph vessels.

SPEC 1.8 The cardiac cycle and the heart

WHAT THIS SPEC POINT ASKS

Know the cardiac cycle (atrial systole, ventricular systole and cardiac diastole) and relate the structure and operation of the mammalian heart, including its major blood vessels, to its function. (Myogenic stimulation isn't needed at AS.)

Heart structure

Diagrams show the heart as if it's facing you, so its **right side is on the left of the page**.

- **Right side**: deoxygenated blood in from the body (**vena cava**) → out to the lungs (**pulmonary artery**).
- **Left side**: oxygenated blood in from the lungs (**pulmonary vein**) → out to the body (**aorta**).
- **Atrioventricular (AV) valves** between atria and ventricles: **tri**cuspid on the **ri**ght, **bi**cuspid (mitral) on the left. **Tendinous cords** stop them turning inside out.
- **Semilunar valves** at the base of the aorta and the pulmonary artery.
- **Left ventricle wall thicker** than the right: it generates a higher pressure to pump blood all round the body. The right only pumps to the nearby lungs, and low pressure there protects the lung capillaries.
- **Atria** have thin walls: they only push blood a short distance, into the ventricles.
- **Septum**: keeps oxygenated and deoxygenated blood separate.
- **Coronary arteries**: supply the heart muscle itself with oxygenated blood.

The three stages (one cycle ≈ 0.8 s at 75 beats per minute)

The pressure rule: **contraction** → **smaller volume** → **higher pressure**. Blood always flows from higher to lower pressure.

1. **Atrial systole**: atria contract → atrial pressure rises above ventricular pressure → blood is pushed through the open **AV valves** into the ventricles. Semilunar valves are closed.
2. **Ventricular systole**: ventricles contract (from the apex, at the bottom, upwards) → ventricular pressure rises above atrial pressure → **AV valves shut** (the "lub" sound). When ventricular pressure rises above the

pressure in the arteries, the **semilunar valves open** and blood is ejected into the aorta and pulmonary artery.

3. **Cardiac diastole**: atria and ventricles relax → ventricular pressure falls below arterial pressure → **semilunar valves shut** (the "dub" sound). Blood flows from the veins into the atria and, once atrial pressure is above ventricular pressure, the **AV valves open** and the ventricles fill passively.

Stage	Atria	Ventricles	AV valves	Semilunar valves
Atrial systole	contract	relaxed	open	closed
Ventricular systole	relaxed	contract	**closed**	**open**
Diastole	relaxed	relaxed	open	closed

Valves open when pressure behind them is higher than pressure in front, and **close** when that difference reverses. They are moved by pressure; they don't contract.

One cardiac cycle, in order

1. Atrial systole: the atria contract and push blood through the open atrioventricular valves into the ventricles.

   Extra filling gives the ventricles a little more blood before they contract.

2. Ventricular systole begins: pressure in the ventricles rises above the atrial pressure.

   The rise in pressure shuts the atrioventricular valves, which stops backflow into the atria.

3. Pressure rises further, until it exceeds the pressure in the aorta and pulmonary artery.

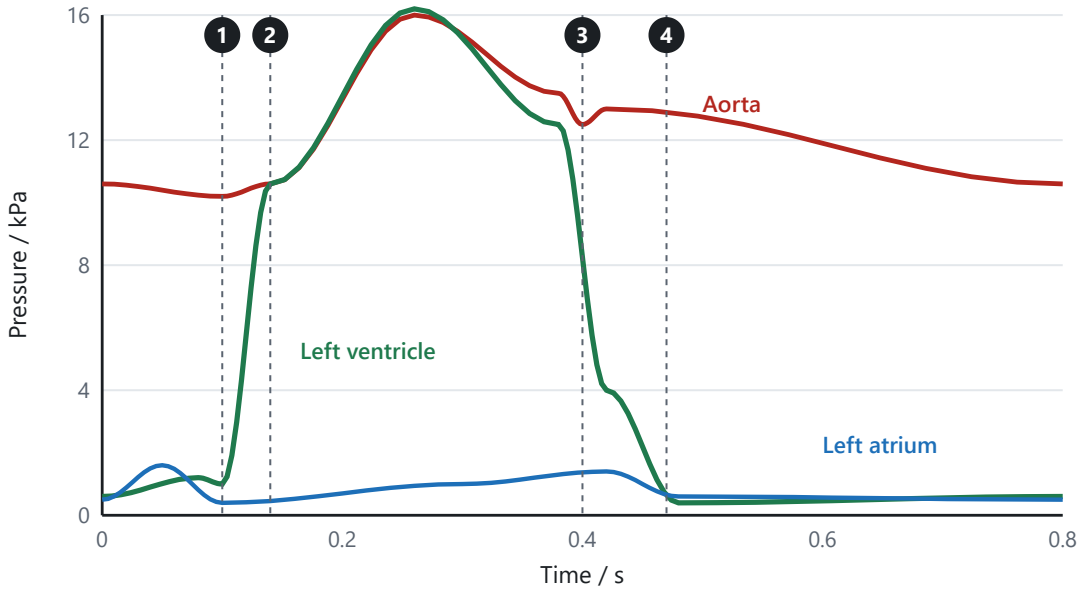
   The semilunar valves are forced open, so blood leaves the heart.

4. Cardiac diastole: the ventricles relax, and pressure in them falls below the pressure in the arteries.

   The semilunar valves snap shut, so blood cannot flow back into the ventricles.

5. When ventricular pressure falls below atrial pressure, the atrioventricular valves open and the ventricles fill passively from the atria.

   The filling is driven by the pressure difference, not by muscle contraction.



Left side of the heart. 1: AV valve closes (ventricle > atrium). 2: semilunar valve opens (ventricle > aorta). 3: semilunar valve closes (aorta > ventricle). 4: AV valve opens (atrium > ventricle).

Reading the pressure graph

- Each point where two lines **cross** is a valve opening or closing.
- From **2** to **3** blood is being ejected into the aorta.
- From **1** to **2** and from **3** to **4**, all valves are shut and the ventricle volume isn't changing.
- The small notch in aortic pressure just after **3** is the semilunar valve snapping shut. Aortic pressure stays high in diastole because the elastic wall **recoils**.
- The right ventricle's curve has the same shape but a much **lower peak** pressure.

WORKED EXAMPLE

Worked example: heart rate from a graph

One complete cycle (peak to the *same* point on the next peak) takes 0.8 s.

Beats per second = 1 ÷ 0.8 = 1.25; beats per minute = 1.25 × 60 = **75 bpm**.

(Shortcut: 60 ÷ cycle time in seconds. Reverse: cycle time = 60 ÷ heart rate.)

Follow the s.f. instruction exactly: to 3 s.f. the answer is **75.0**, and "75" loses the mark.

MARK-SCHEME WORDING

- Valve states: atrial systole AV open, SL closed; ventricular systole AV closed, SL open; cardiac diastole AV open, SL closed. The AV valves close at the start of ventricular systole.

- Call it "(cardiac) diastole", not "ventricular diastole" or "diasystole".
- Coronary artery: brings "oxygenated blood to the heart muscle" "for (aerobic) respiration". It branches off the aorta because that's the nearest vessel carrying oxygenated blood. Glucose / nutrients are ignored.
- Exercise: "the cardiac cycle is shorter / completed more frequently", "ventricles contract more forcefully". "Heart rate increases" alone is ignored.
- Right v left ventricle: same timing (both sides "beat simultaneously"); the right peaks lower because it pumps only to the lungs, so the alveoli aren't damaged.
- Delay between atrial and ventricular contraction: so the "atria can finish emptying" into the ventricles before the AV valves close.
- Volume graphs: left ventricle volume increases in atrial systole, decreases in ventricular systole, increases in diastole.

Often asked as: valve-state MCQs and cloze passages, label vessels or draw flow arrows, read pressure or volume graphs, calculate heart rate or cycle length.

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Examined in past papers even though the spec doesn't list it.

Cardiac output = stroke volume × heart rate (e.g. 70 cm^3 × 75 min^{-1} = 5250 cm^3 min^{-1}). Formally later content, but Unit 1 data questions use it.

Delay at the AV node: the impulse is held up briefly before it passes down the septum, so the atria empty before the ventricles contract. The spec leaves this out at AS, but it has been rewarded (WBI11 Oct 2022).

Pulmonary embolism: a clot lodged in a pulmonary artery stops blood being oxygenated → breathlessness and a raised heart rate (WBI11 Jan 2025).

COMMON MISTAKE

Mixing up pressure and volume, or forgetting to compare. Every valve explanation needs a comparison: "pressure in the ventricle is **higher than** in the atrium, so the AV valve closes."

Swapping the pulmonary artery and pulmonary vein. Thinking the right ventricle contracts after the left: they contract together.

Writing "heart rate increases" when the question asks about events in the cycle. Working out 73 ÷ 60 when you need 60 ÷ 73.

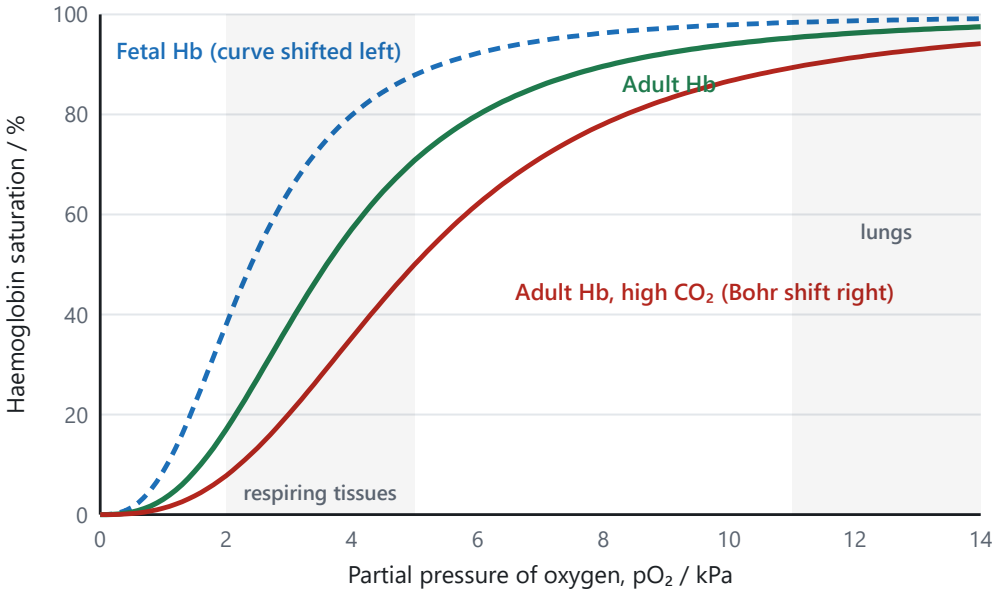
SPEC 1.9 Haemoglobin: carrying O_2 and CO_2

WHAT THIS SPEC POINT ASKS

(i) Understand the role of haemoglobin in transporting oxygen and carbon dioxide. (ii) Understand the oxygen dissociation curve, the Bohr effect, and why fetal haemoglobin's higher affinity for oxygen matters.

Haemoglobin and oxygen

- Haemoglobin (Hb) is a **globular protein** with **four polypeptide chains** (2 α, 2 β). Each chain holds a **haem group** containing **Fe^{2+}**, which binds one O_2 molecule. So one Hb carries **4 O_2** (8 oxygen atoms).
- Hb + $4O_2$ ⇌ oxyhaemoglobin. **Loading** (association) happens in the lungs; **unloading** (dissociation) in respiring tissues.
- **Cooperative binding**: the first O_2 binds with difficulty. Once it binds, Hb undergoes a **conformational change** (its tertiary/quaternary structure changes), making it easier for further O_2 molecules to bind. This is what makes the curve **S-shaped** (sigmoid).



Shift to the right = lower affinity (O_2 released more readily). Shift to the left = higher affinity (O_2 taken up more readily).

Reading the curve, in both directions

- **Left → right (loading):** at low pO_2 the curve is shallow: the first O_2 binds with difficulty. In the steep middle, binding is easy after the shape change, so a small rise in pO_2 gives a big rise in saturation. At high pO_2 (lungs) it levels off: Hb is nearly **saturated**, with few free sites.

- **Right → left (unloading):** in the lungs there's little dissociation. In the **steep part**, at the pO_2 of respiring tissues, a small drop in pO_2 releases a **lot** of O_2, exactly where cells need it.
- **Affinity** = how readily Hb takes up O_2. High affinity: loads easily, unloads reluctantly. Low affinity: the opposite.

The Bohr effect (shift to the right)

1. Respiring tissue produces **CO_2**.
2. In red blood cells, $CO_2 + H_2O \rightarrow H_2CO_3$ (carbonic acid), catalysed by **carbonic anhydrase**. $H_2CO_3 \rightarrow$ **H^+ + HCO_3^-**.
3. The H^+ ions (lower pH) bind to Hb and change its shape, **lowering its affinity** for O_2.
4. So **more O_2 is released** at the same pO_2: the curve shifts **right**. Active tissues make the most CO_2, so they get the most O_2.

Carbon dioxide transport

- About **5%** dissolved in the plasma.
- About **10–20%** bound to Hb as **carbaminohaemoglobin**. The CO_2 joins the **amino groups of the polypeptide chains**, not the haem.
- Most (about **70–85%**) as **hydrogencarbonate ions (HCO_3^-)**: formed in red blood cells as above, then diffuse out into the plasma. Hb picks up the H^+, **buffering** the blood pH. The reaction happens in red cells rather than plasma because that's where the carbonic anhydrase is.

Converting carbon dioxide to hydrogencarbonate in a red blood cell

1. Carbon dioxide diffuses from the tissue fluid into the red blood cell.

   The concentration of carbon dioxide is higher outside the cell, so it moves in down the gradient.

2. Carbon dioxide combines with water to form carbonic acid, H_2CO_3.

   The reaction is catalysed by the enzyme carbonic anhydrase inside the red blood cell, making it very fast.

3. Carbonic acid dissociates into hydrogen ions and hydrogencarbonate ions.

4. Hydrogencarbonate ions diffuse out into the plasma, and chloride ions move into the cell to balance the charge.

   This exchange is called the chloride shift, and it keeps the electrical charge balanced.

5. The hydrogen ions bind to haemoglobin, which helps release oxygen (the Bohr effect).

Fetal haemoglobin (shift to the left)

- Fetal Hb has a **higher affinity** for O_2 than adult Hb.
- At the **placenta** the pO_2 is low. **At the same pO_2**, the mother's Hb unloads O_2 and the fetal Hb, with its higher affinity, **loads** it, so oxygen diffuses from mother to fetus.
- After birth, adult Hb gradually replaces fetal Hb, because the baby now needs Hb that **unloads** O_2 readily in its own respiring tissues.

EXAM TIP

For any curve that's shifted left or right, give three steps: (1) **at the same pO_2**, saturation is higher or lower; (2) so affinity is higher or lower; (3) so O_2 is loaded or unloaded more readily: say *where* and *why that helps*.

MARK-SCHEME WORDING

- Sigmoid curve: "four subunits"; "binding of the first oxygen molecule is difficult"; then "a conformational change" makes "binding of the other molecules easier" (cooperative binding accepted); the curve levels off as Hb becomes saturated. Plain "change of shape" is ignored.
- Structure: 4 polypeptide subunits (2α, 2β), four haem groups, each with an "**iron ion / Fe^{2+}**" (not "iron") that binds "an oxygen molecule / O_2" (not "O"). "Stores oxygen" is rejected. Hydrophilic R groups on the outside make Hb soluble in the red cell's cytoplasm.
- Partial pressure: "the pressure exerted by one gas in a mixture of gases". Not amount or %.
- Fetal Hb: "higher affinity", so "at the same partial pressure" oxygen "leaves the adult haemoglobin and binds to fetal haemoglobin". Write fetus, not "baby".
- Left shift: "higher affinity", "higher % saturation at a lower pO_2", enough O_2 for respiration. Right shift: "lower affinity", so O_2 is unloaded more readily to respiring tissues.
- Altitude: lower pO_2 in the alveoli → smaller gradient → **slower rate of diffusion** → Hb less saturated. "Less diffusion" is ignored.

Often asked as: explain the sigmoid shape, link Hb's structure to its function, explain a shifted curve in an unfamiliar animal (often a 6-mark), read values off a curve.

COMMON MISTAKE

"CO_2 binds to the haem and pushes the oxygen off." It's the **H^+ ions** (from carbonic acid) binding to Hb that lower its affinity. Say "**partial pressure** of oxygen", not "amount of oxygen".

Thinking fetal Hb works because fetal blood has a lower pO_2. At the placenta both bloods are at about the same pO_2: the point is fetal Hb's **higher affinity**.

Writing "iron" for **iron ion (Fe^{2+})**, "speed of dissociation" for **affinity**, or "moves" for **diffuses**. Confusing Hb (the protein) with the red blood cell. Quote values read off the curve.

In multi-animal 6-mark questions, comparing traits across the animals scores; describing one species at a time stays at Level 1.

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Fetal Hb has **γ (gamma) chains** in place of the β chains (WBI11 Oct 2024).

Data questions use other animals. Animals living where O_2 is scarce (high altitude, burrows, mud such as lugworms) have curves **left** of adult human Hb (higher affinity). Small, very active mammals have curves to the **right** (unload readily). **Myoglobin** (one chain, an O_2 store in muscle) has a curve far to the left: it releases O_2 only when pO_2 is very low (WBI11 Jun 2024).

Quick checks

QUICK CHECK

Why is the wall of the left ventricle thicker than the wall of the right ventricle?

- It has to hold more blood than the right ventricle
- It contains more valves
- It must generate higher pressure to pump blood round the whole systemic circuit
- It needs more oxygen than the lungs

The left ventricle pumps blood through the high-resistance systemic circuit, so it needs to generate high pressure. The right ventricle pumps only to the nearby lungs, where lower pressure protects the capillaries. The commonest wrong answer is about volume; both ventricles pump roughly the same volume per beat.

QUICK CHECK

During ventricular systole, which valves are open?

- Both atrioventricular valves
- Neither set of valves
- The atrioventricular valves only
- The semilunar valves

In systole, pressure in the ventricles rises above the pressure in the arteries, which forces the semilunar valves open. The atrioventricular valves are shut to stop backflow into the atria. Students often assume the valves that open during filling are open at the same time as the ones pushing blood out.

QUICK CHECK

In active muscle, the oxygen dissociation curve shifts to the right. What does this mean for oxygen delivery?

- Haemoglobin binds oxygen more tightly
- Haemoglobin binds oxygen less tightly, so more oxygen is released to the tissue
- Haemoglobin becomes more saturated at all partial pressures
- The lungs load less oxygen

A right shift means lower affinity: at any oxygen partial pressure less oxygen is bound, so more is released where it is needed. The commonest error is to read a right shift as higher affinity, which is the opposite of what happens.

QUICK CHECK

What is the main role of the enzyme carbonic anhydrase in red blood cells?

- It binds oxygen to haemoglobin
- It breaks down haemoglobin
- It transports chloride ions out of the cell
- It speeds up the reaction between carbon dioxide and water

Carbonic anhydrase catalyses the formation of carbonic acid from carbon dioxide and water in the tissues, and the reverse reaction in the lungs. Chloride shift is a passive exchange, not carried out by this enzyme.